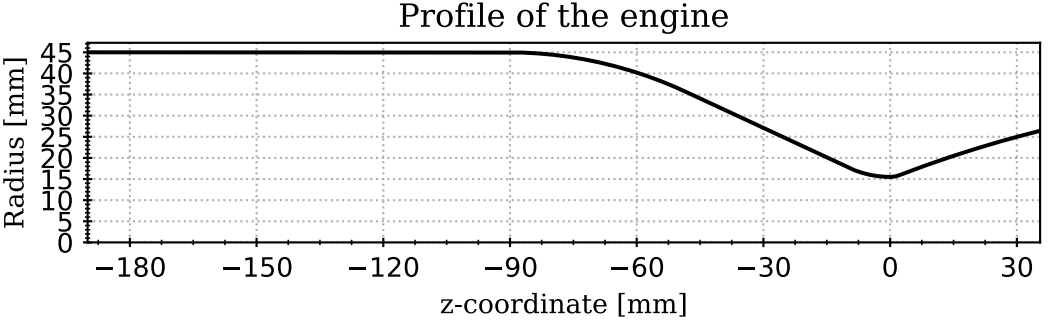
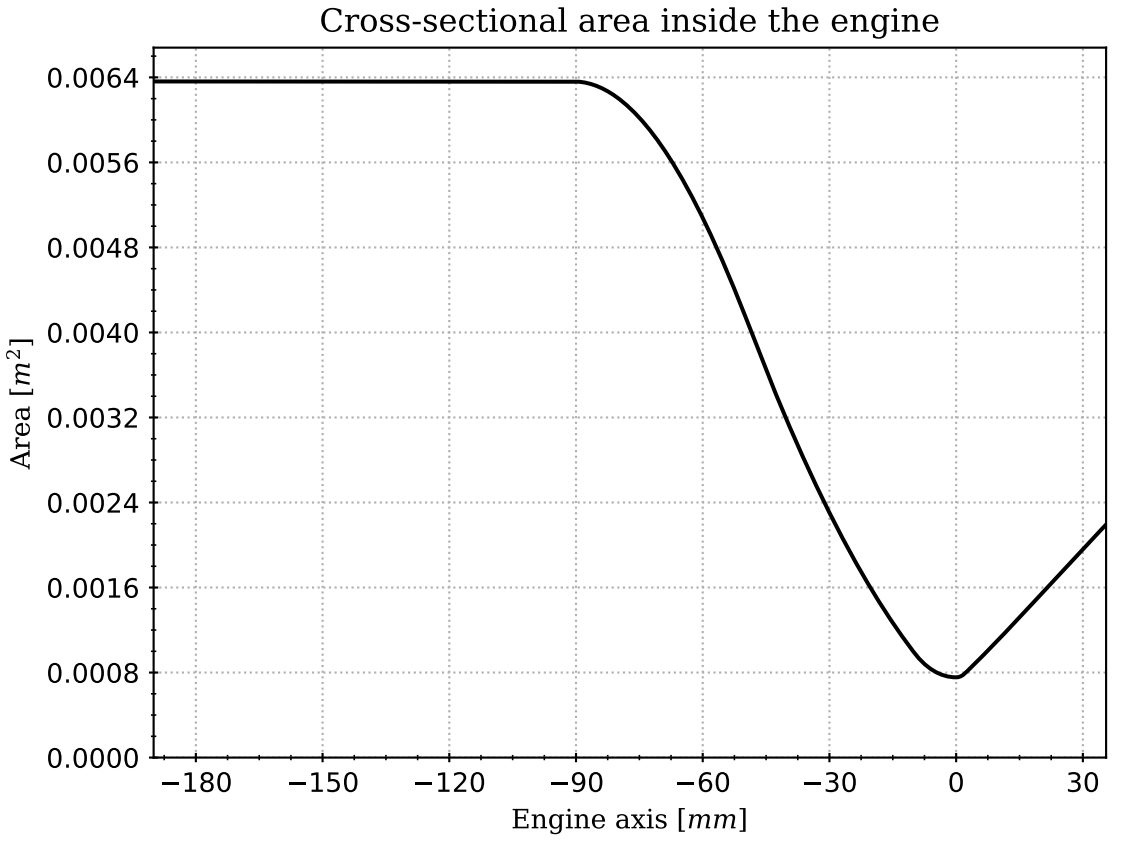
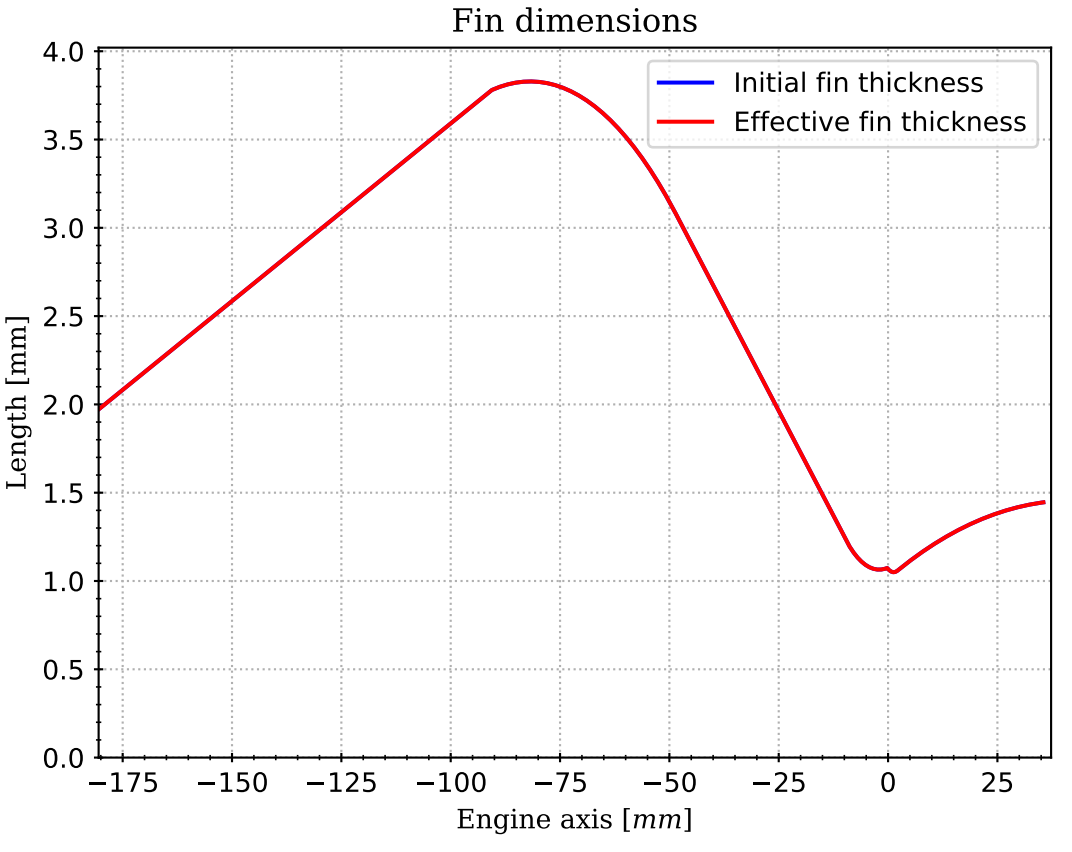
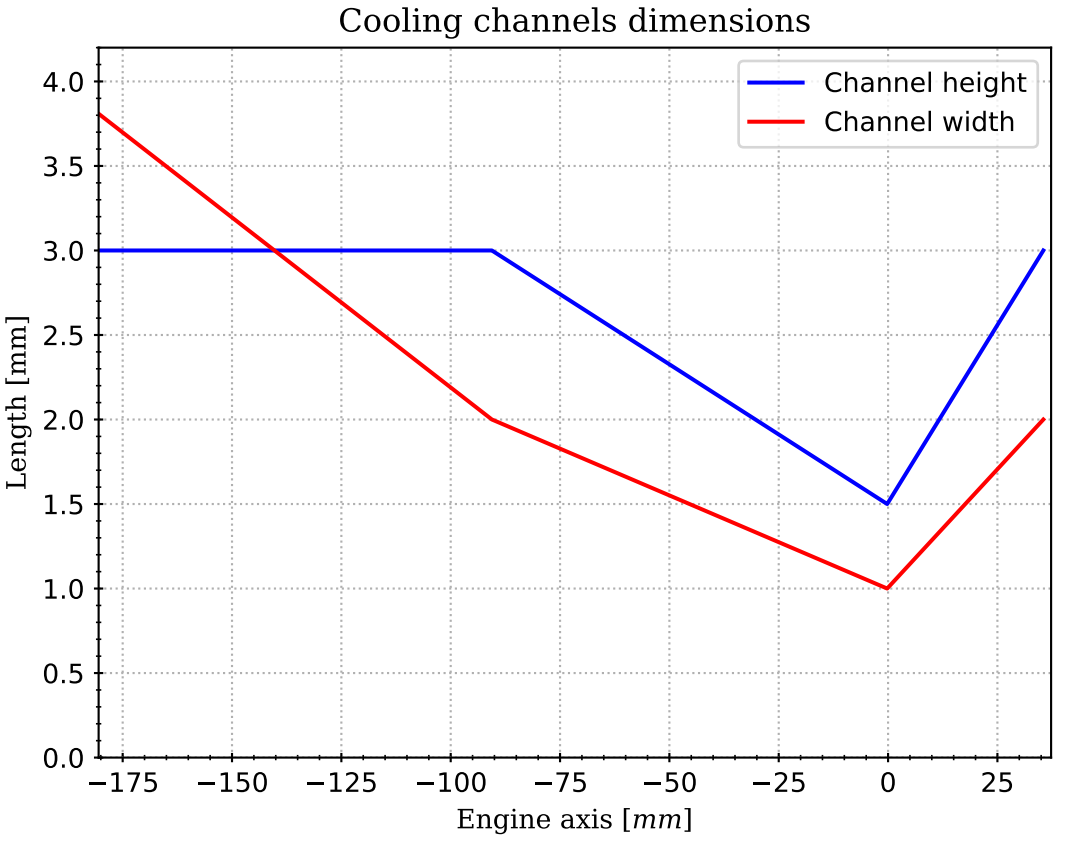
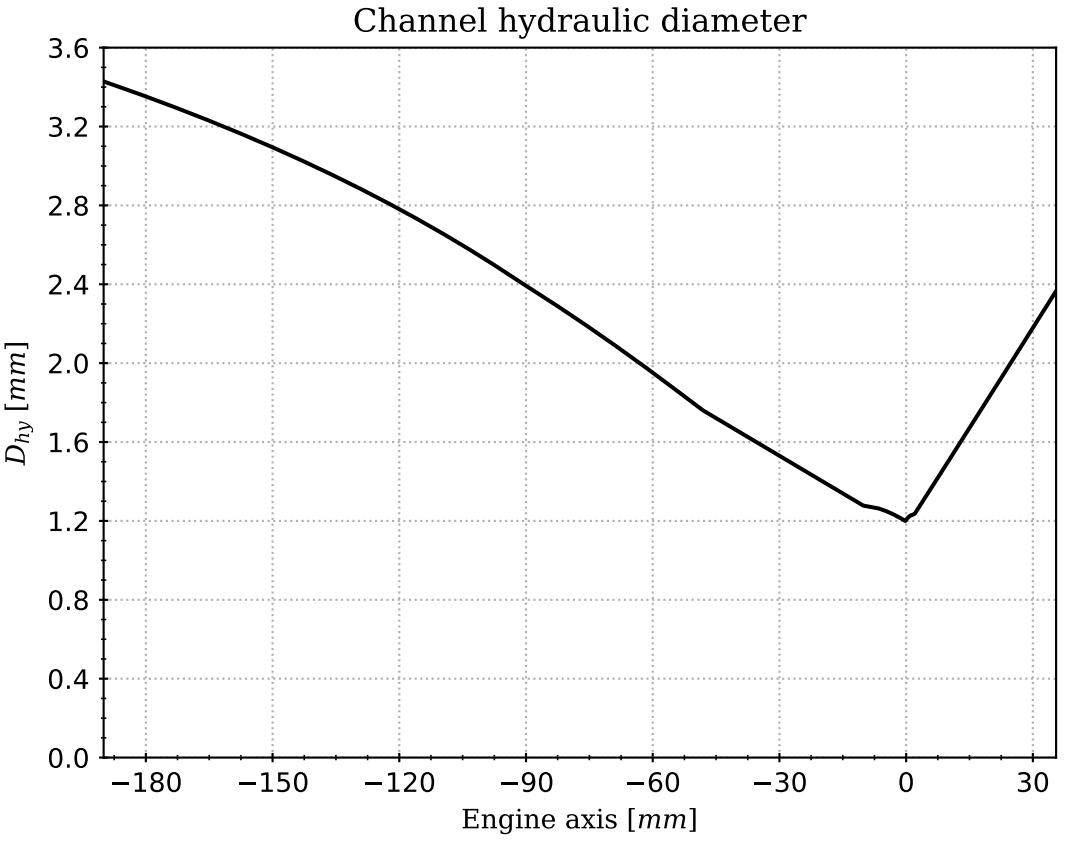
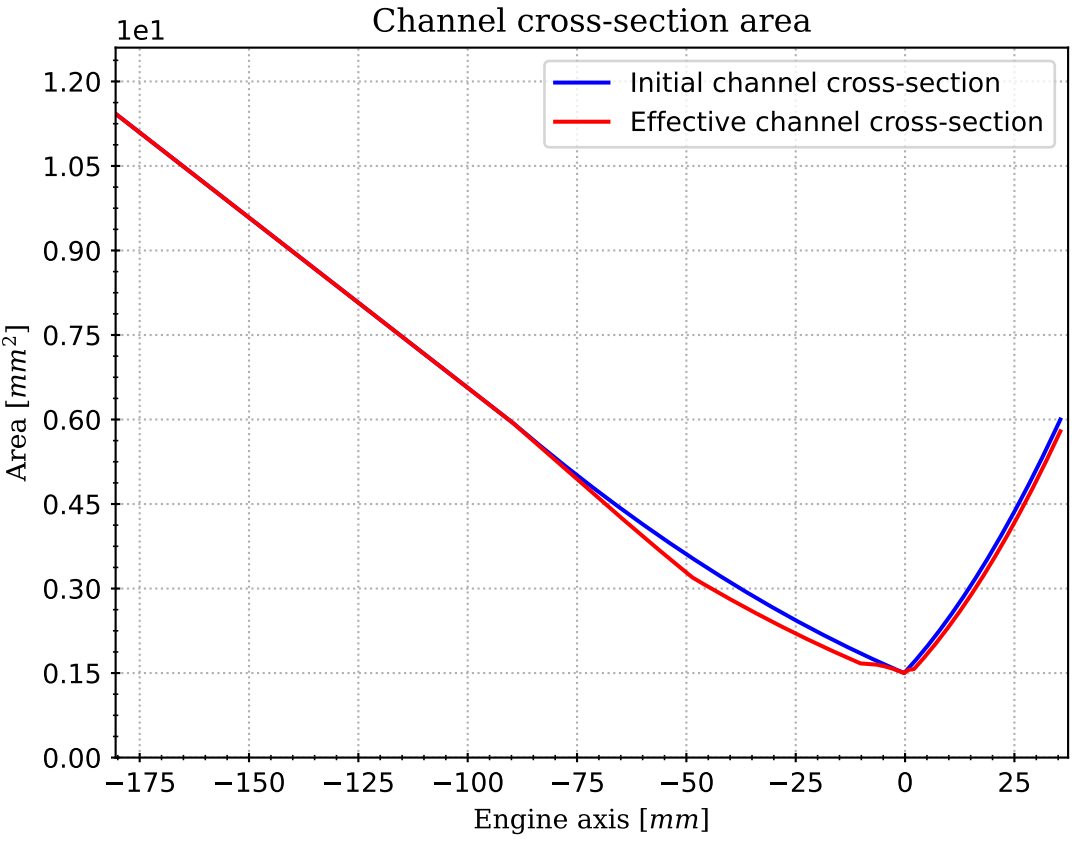

Profile of the engine
Radius [mm]
45
40
35
30
25
20
15
10
5
0
−180
−150
−120
−90
−60
−30
0
30
z-coordinate [mm]

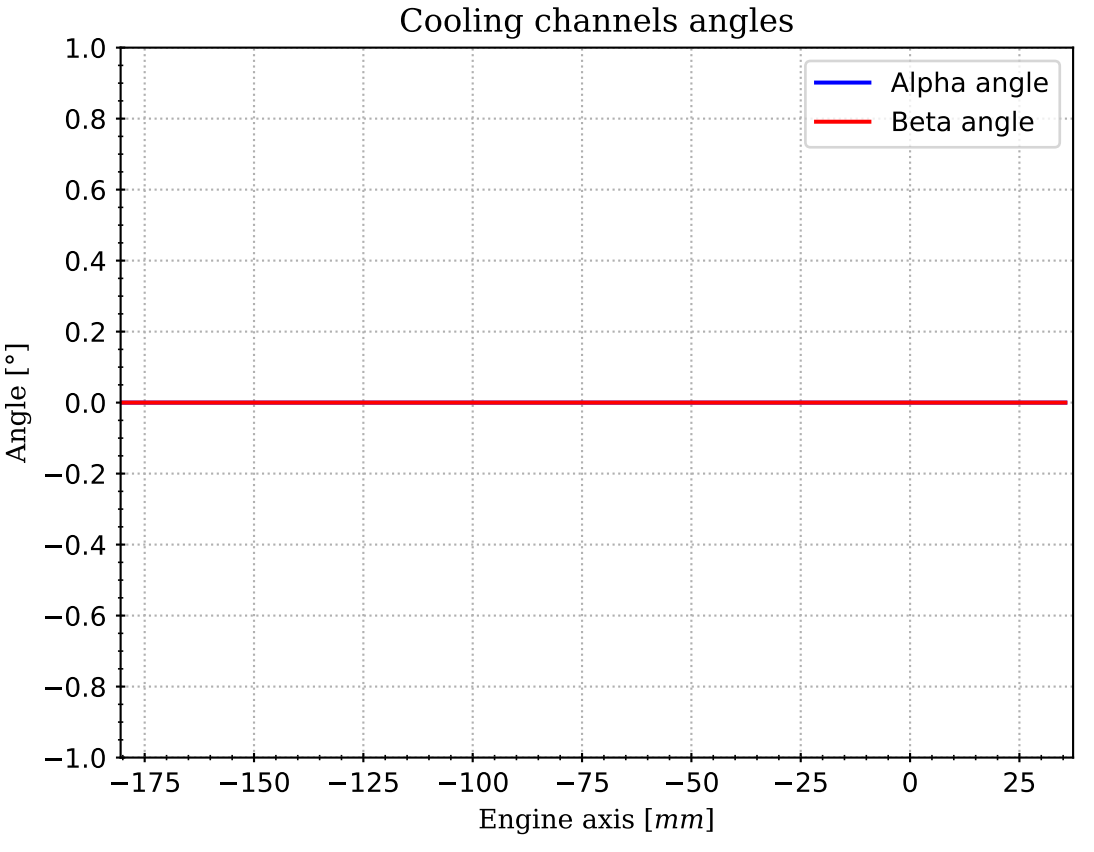

Cross-sectional area inside the engine
0.0064
0.0056
0.0048
0.0040
0.0032
0.0024
0.0016
0.0008
0.0000
Area [m^2]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

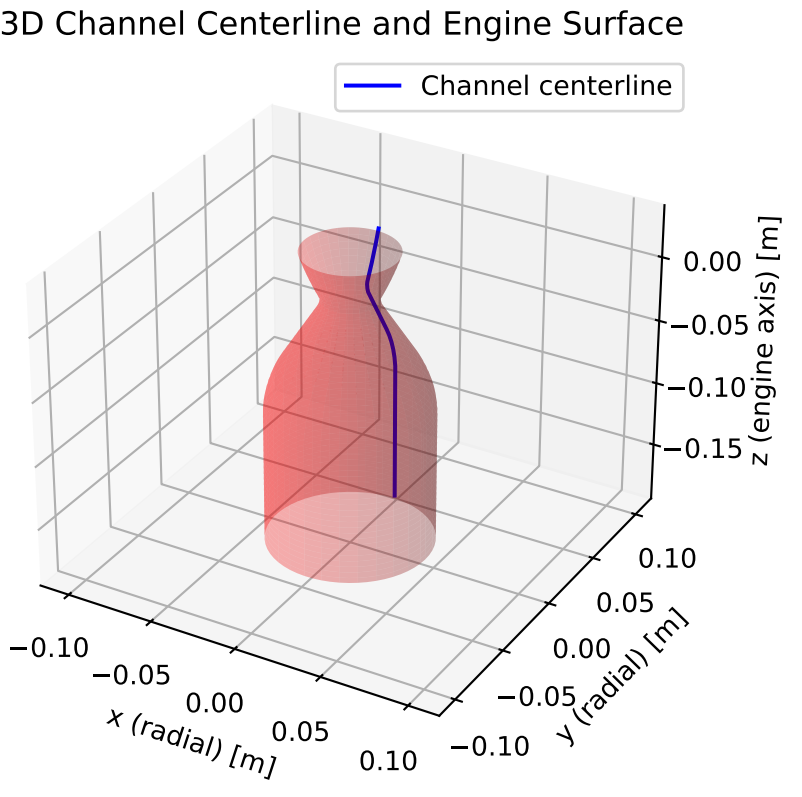

Fin dimensions
Initial fin thickness
Effective fin thickness
Length [mm]
Engine axis [mm]
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−175
−150
−125
−100
−75
−50
−25
0
25

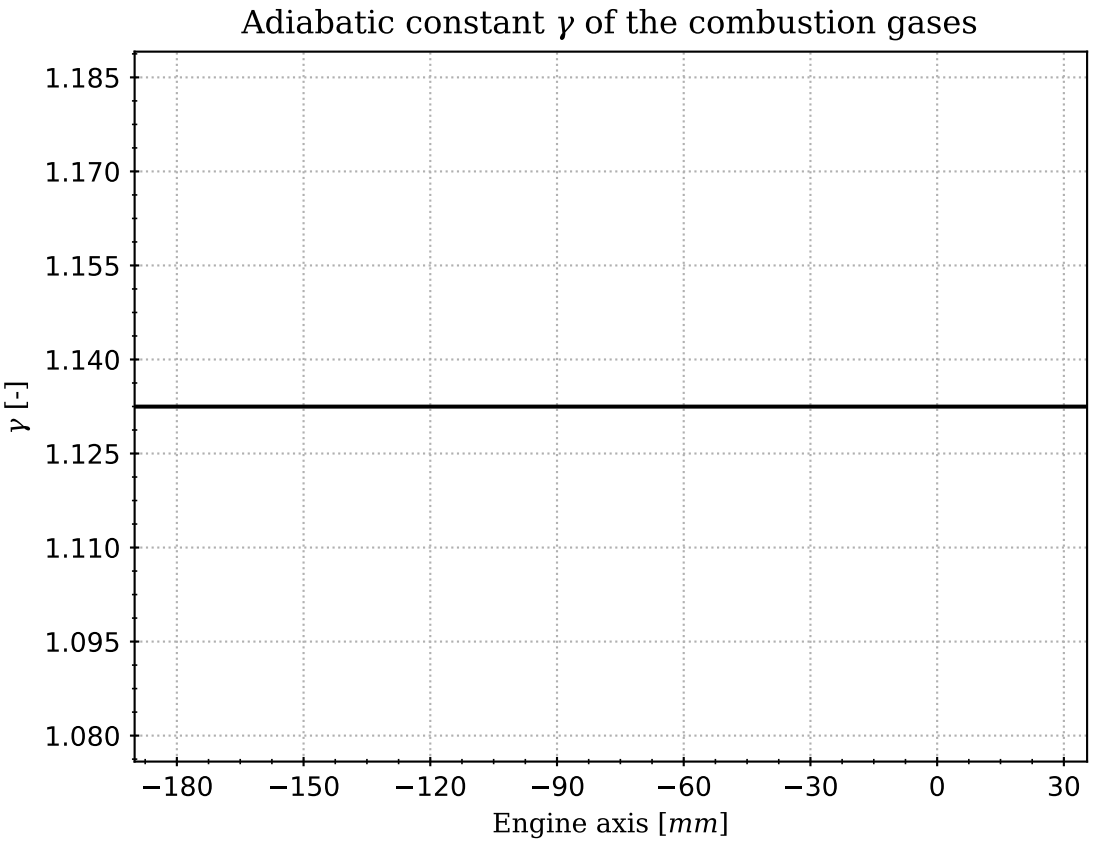

Cooling channels dimensions
Channel height
Channel width
Length [mm]
Engine axis [mm]
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−175
−150
−125
−100
−75
−50
−25
0
25

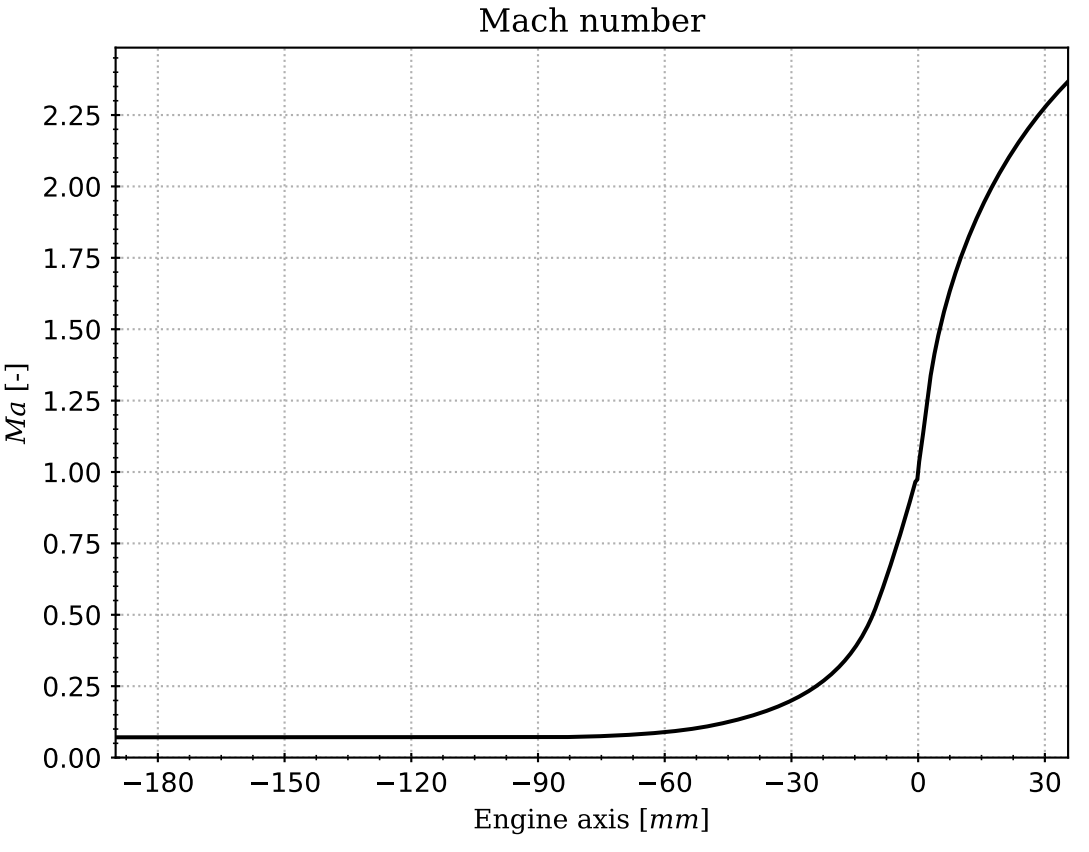

Channel hydraulic diameter
3.6
3.2
2.8
2.4
2.0
1.6
1.2
0.8
0.4
0.0
D_{hy} [mm]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

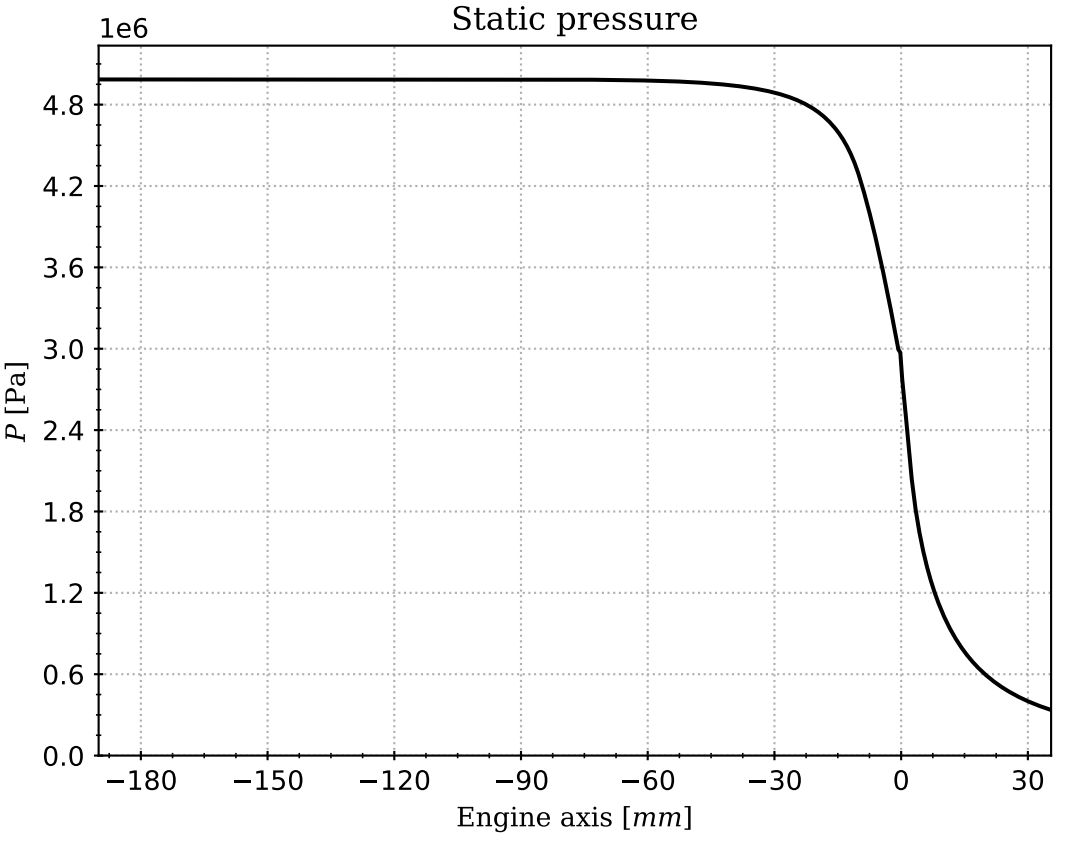

Channel cross-section area
1e1
1.20
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
Area [mm^2]
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]
Initial channel cross-section
Effective channel cross-section

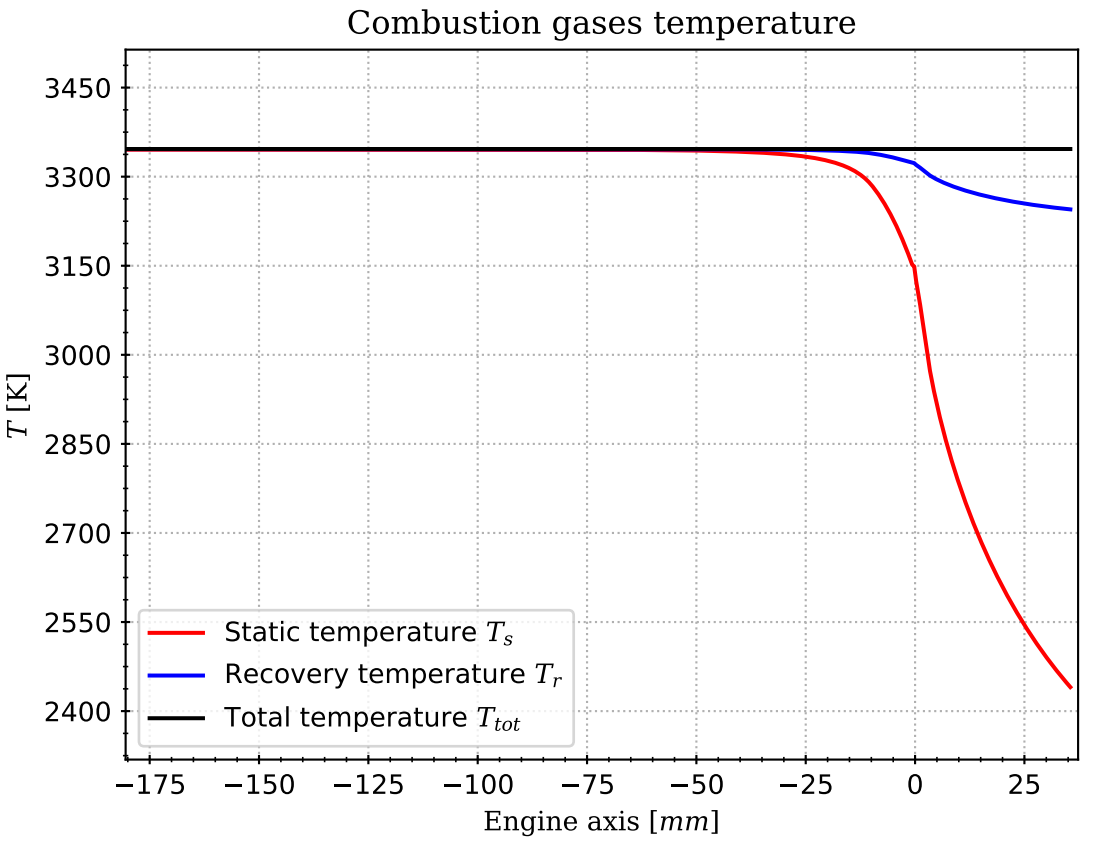

Cooling channels angles
Alpha angle
Beta angle
Angle [°]
Engine axis [mm]
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−175
−150
−125
−100
−75
−50
−25
0
25

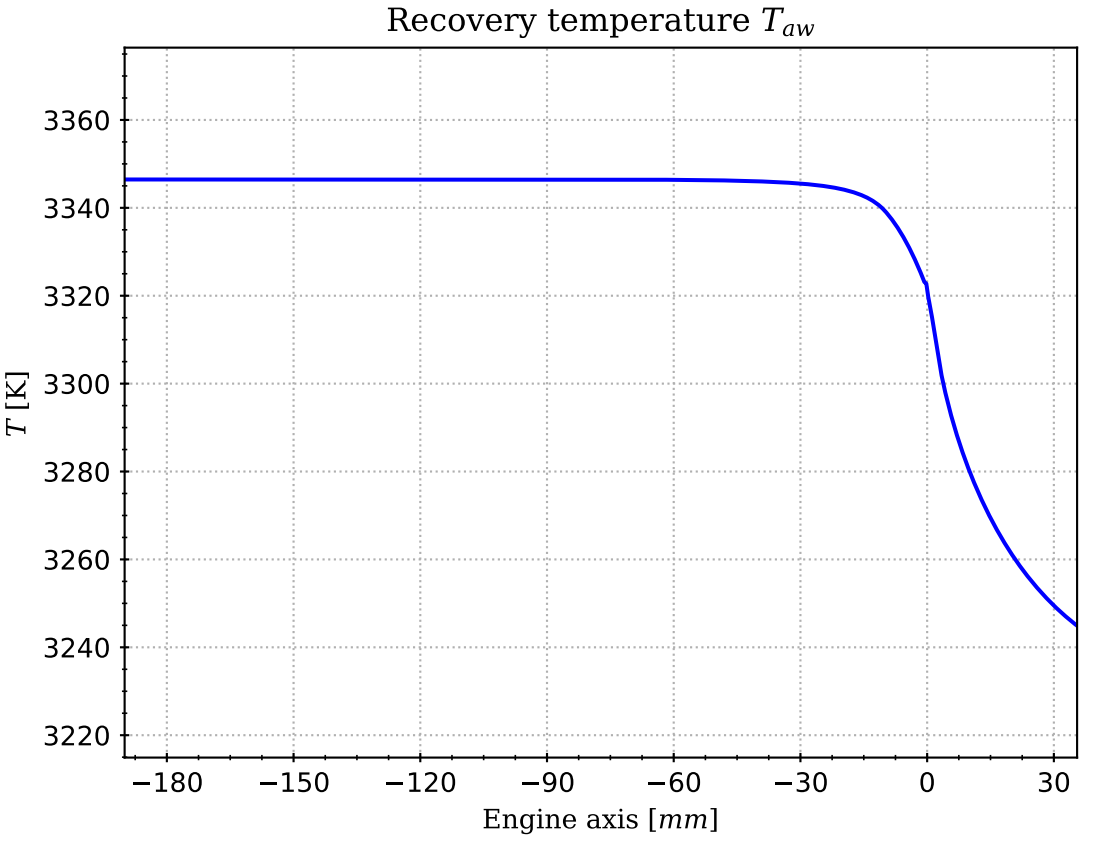

3D Channel Centerline and Engine Surface
Channel centerline
0.00
−0.05
−0.10
−0.15
z (engine axis) [m]
0.10
0.05
0.00
−0.05
−0.10
y (radial) [m]
−0.10
−0.05
0.00
0.05
0.10
x (radial) [m]

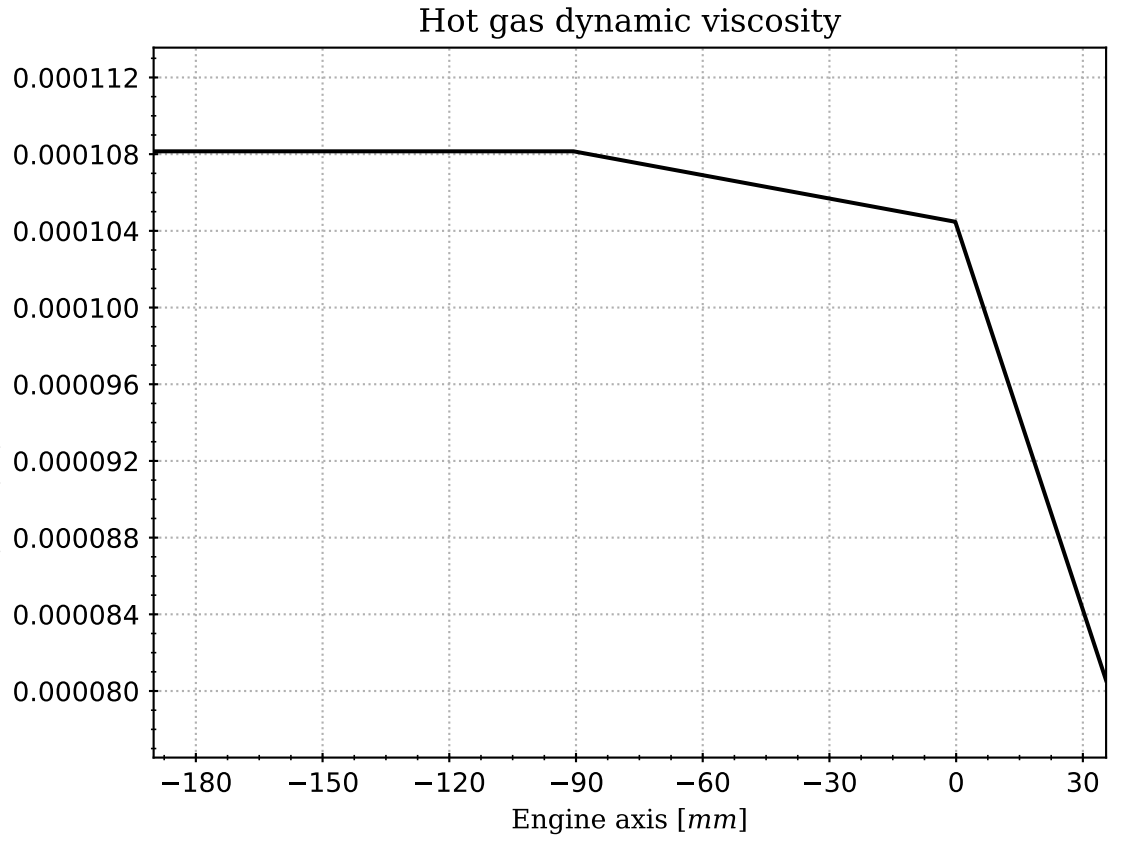

Adiabatic constant γ of the combustion gases
1.185
1.170
1.155
1.140
1.125
1.110
1.095
1.080
γ [-]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

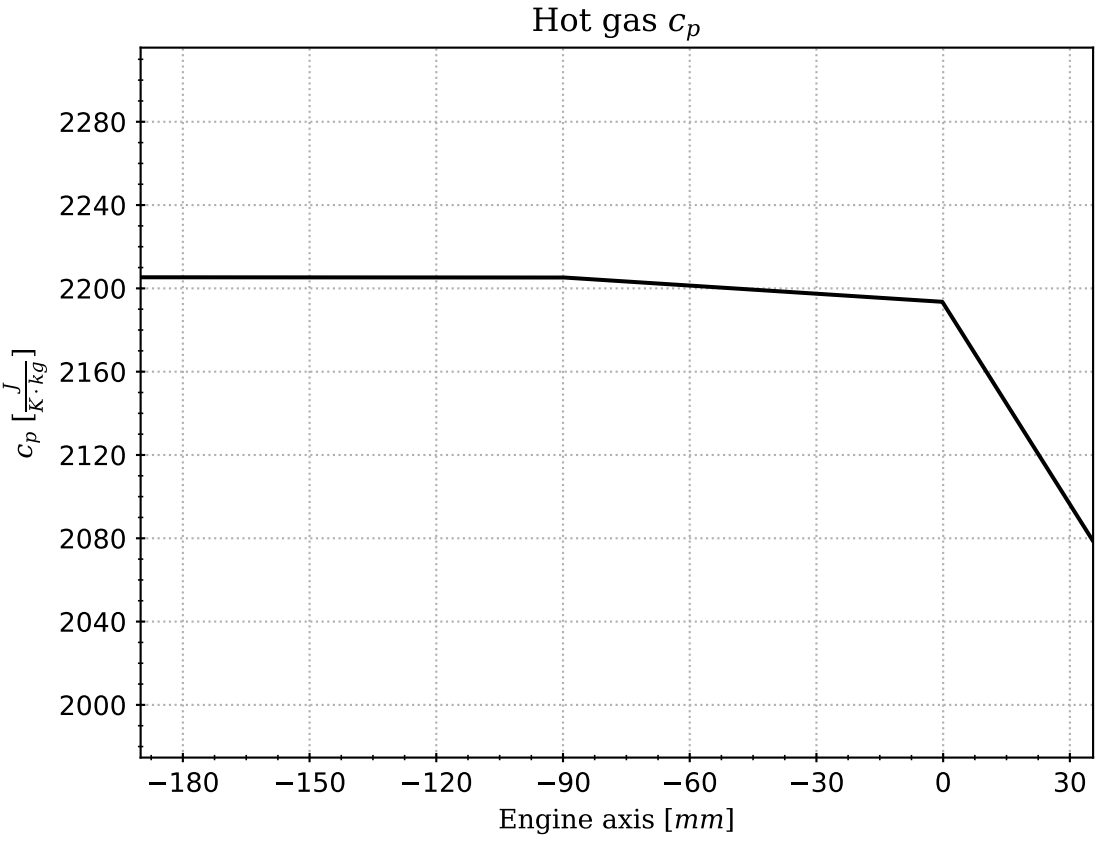

Mach number
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Ma [-]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

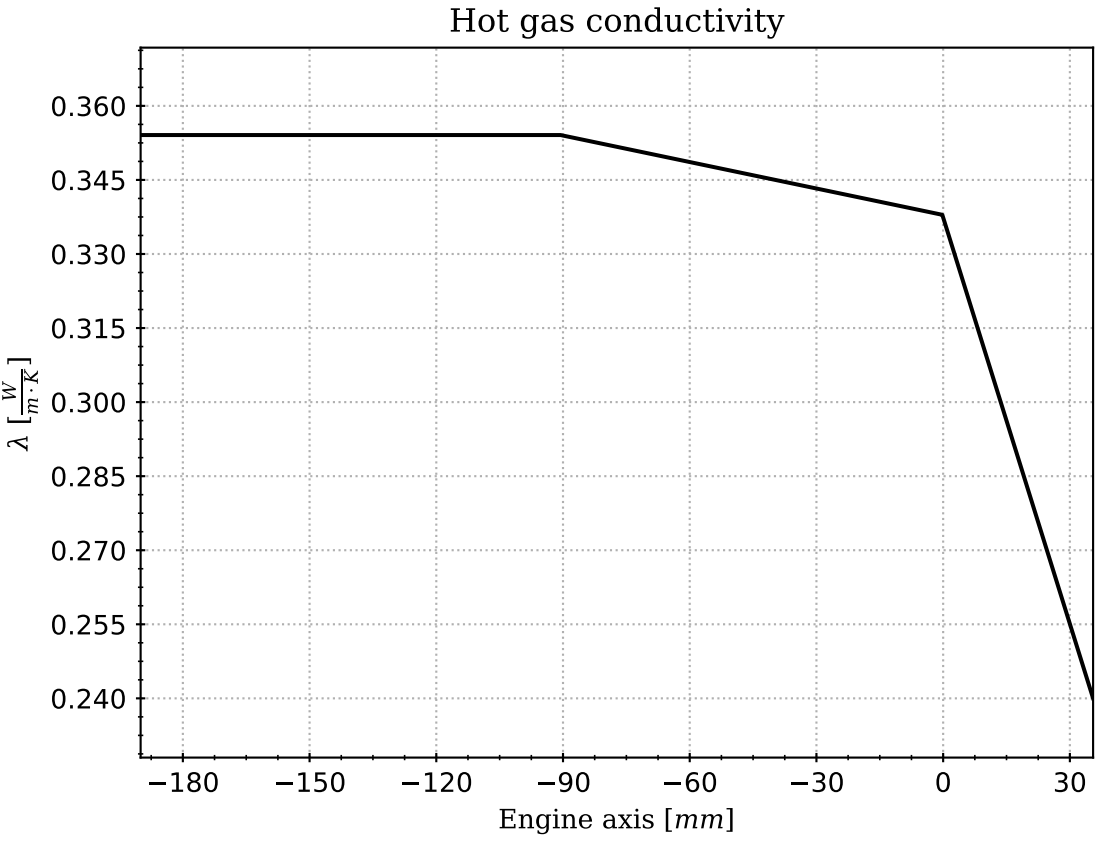

Static pressure
1e6
4.8
4.2
3.6
3.0
2.4
1.8
1.2
0.6
0.0
P [Pa]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

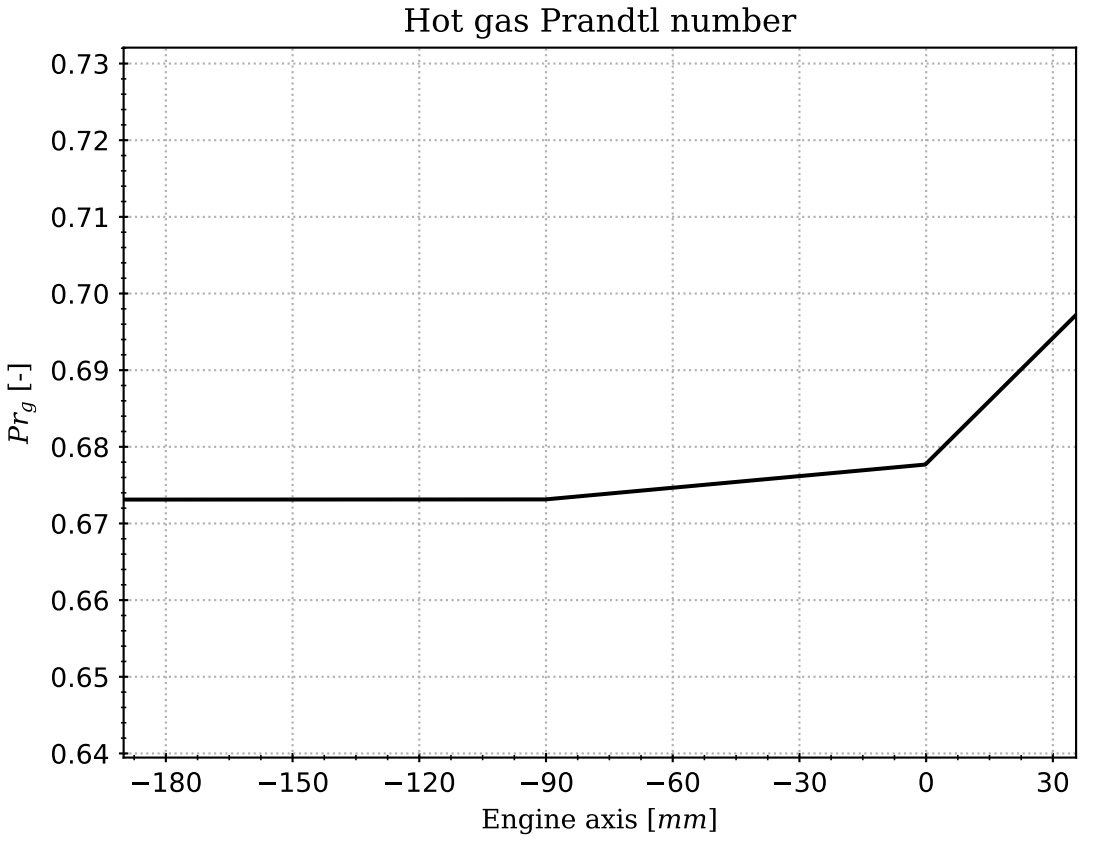

Combustion gases temperature
3450
3300
3150
3000
2850
2700
2550
2400
T [K]
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]
Static temperature T_s
Recovery temperature T_r
Total temperature T_{tot}

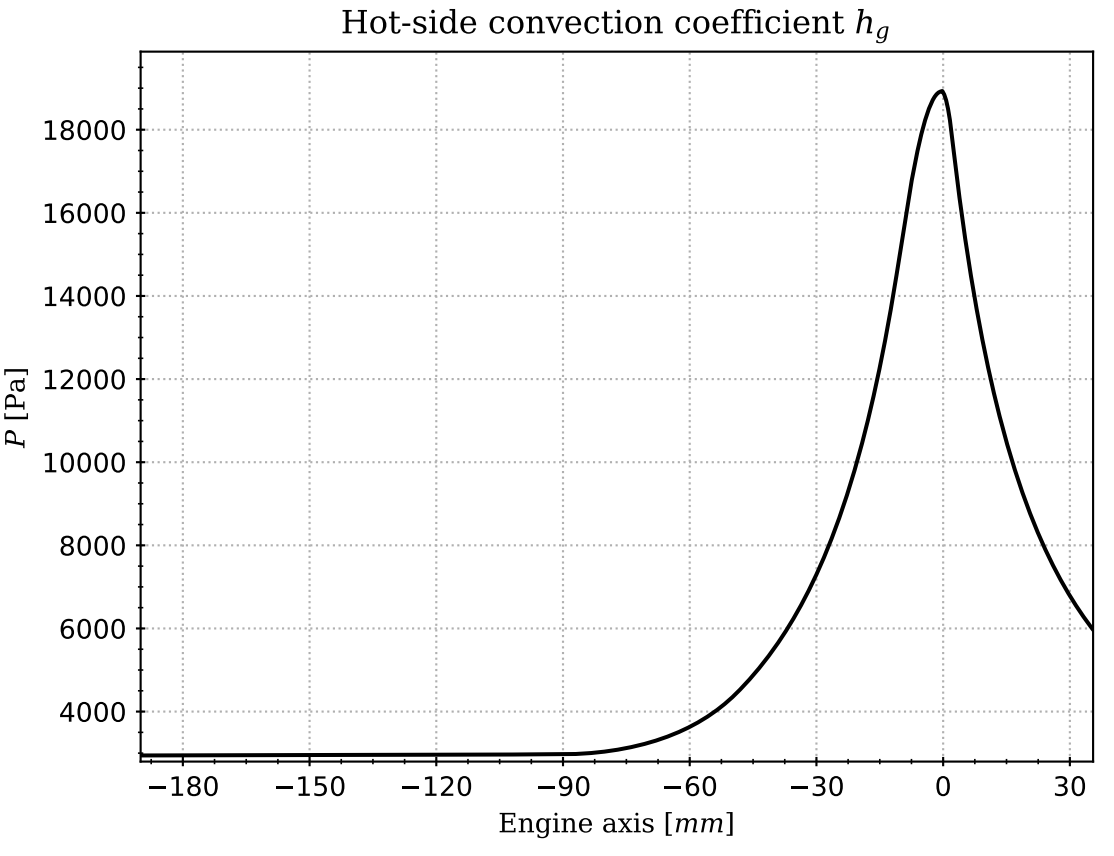

Recovery temperature T_{aw}
3360
3340
3320
3300
3280
3260
3240
3220
T [K]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

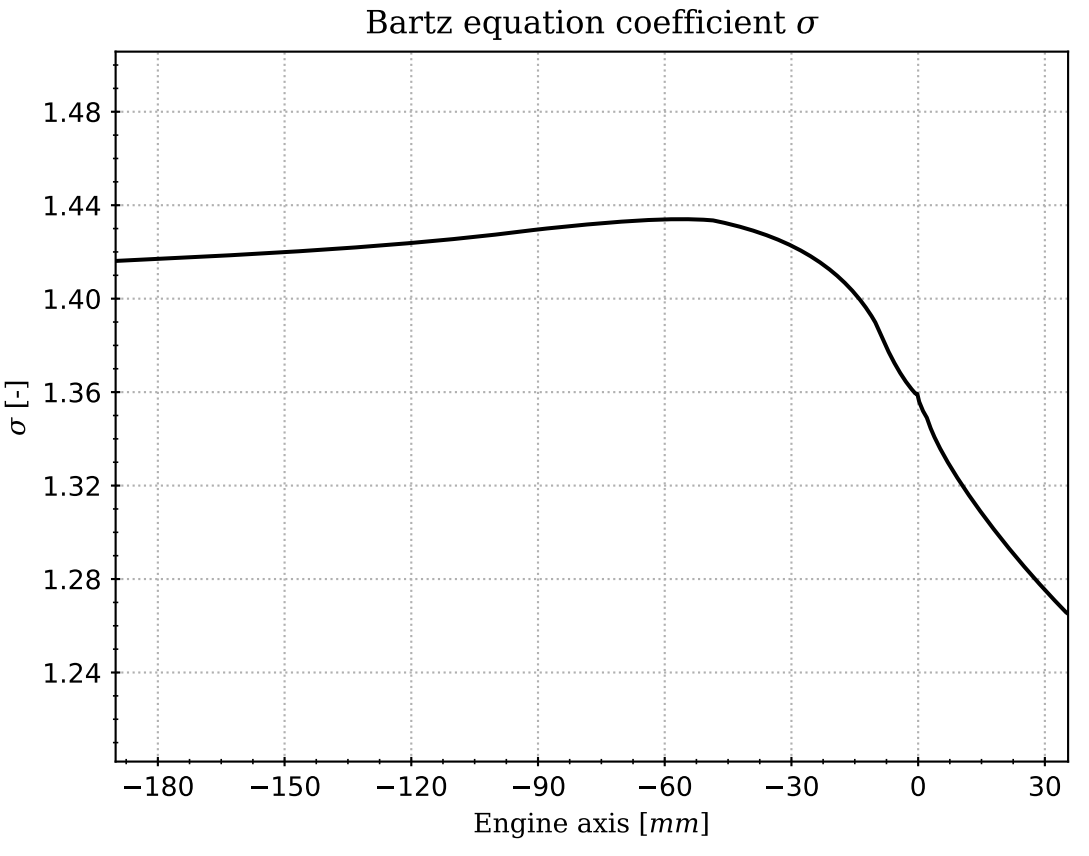

Hot gas dynamic viscosity
0.000112
0.000108
0.000104
0.000100
0.000096
0.000092
0.000088
0.000084
0.000080
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

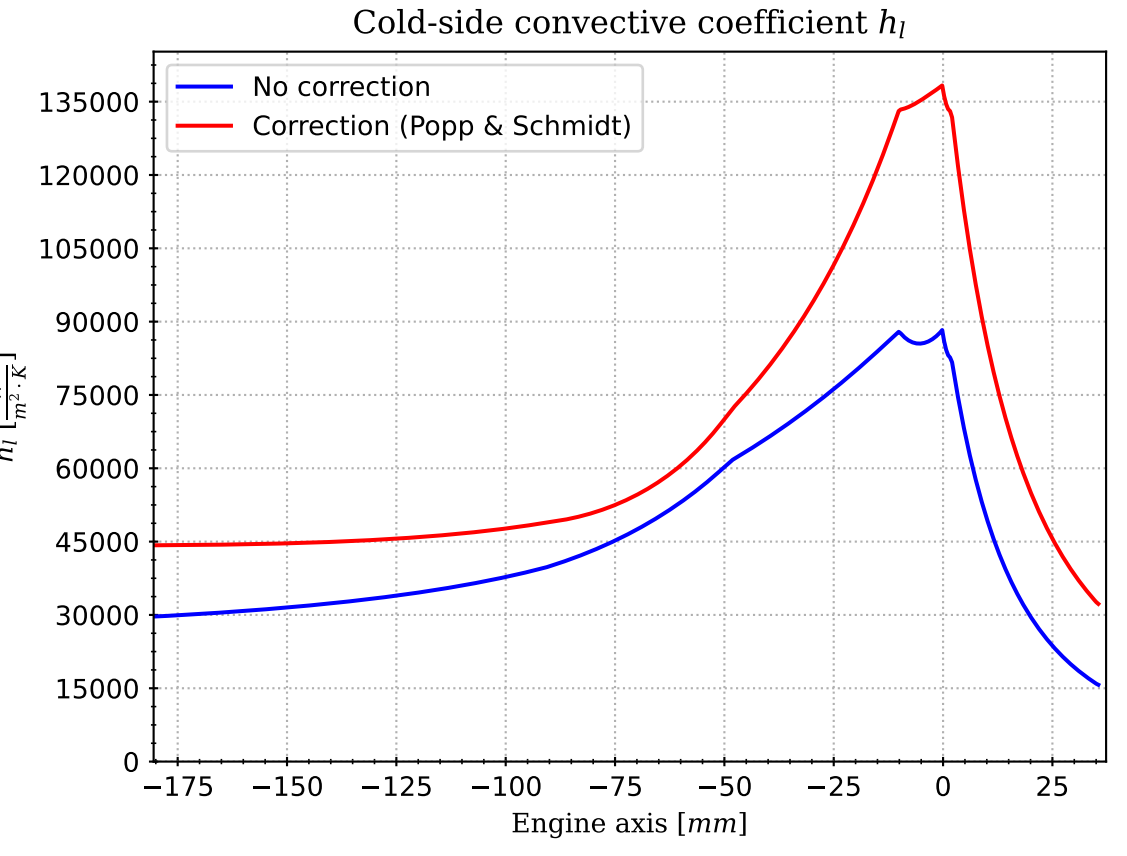

Hot gas c_p
2280
2240
2200
2160
2120
2080
2040
2000
c_p $[\frac{J}{K \cdot kg}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis $[mm]$

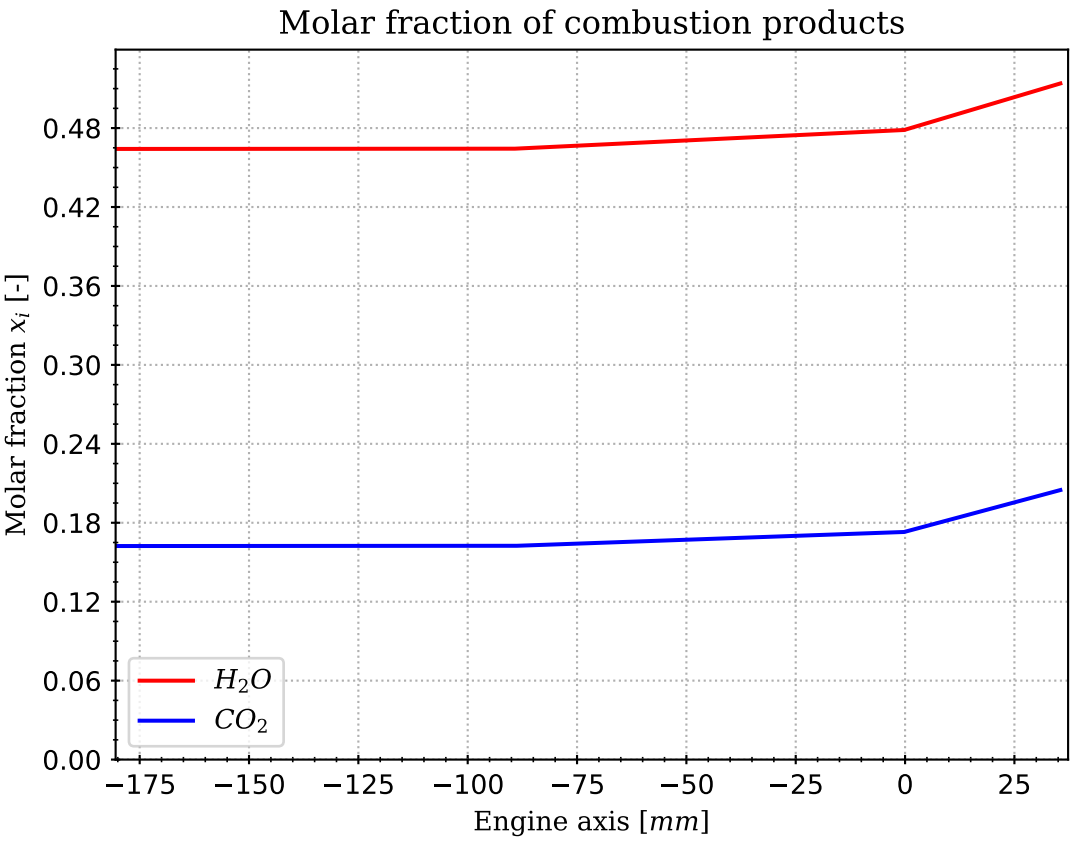

Hot gas conductivity
0.360
0.345
0.330
0.315
0.300
0.285
0.270
0.255
0.240
$\lambda\ [\frac{W}{m \cdot K}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

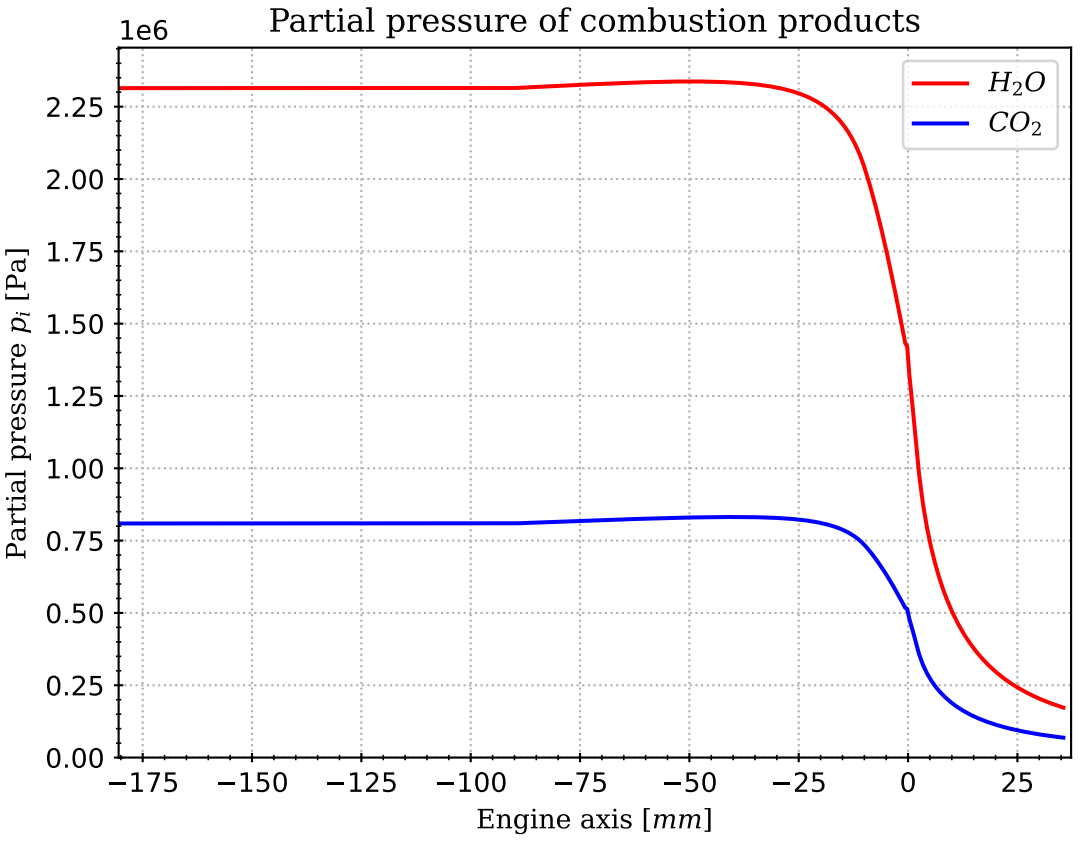

Hot gas Prandtl number
0.73
0.72
0.71
0.70
0.69
0.68
0.67
0.66
0.65
0.64
Pr_g [-]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

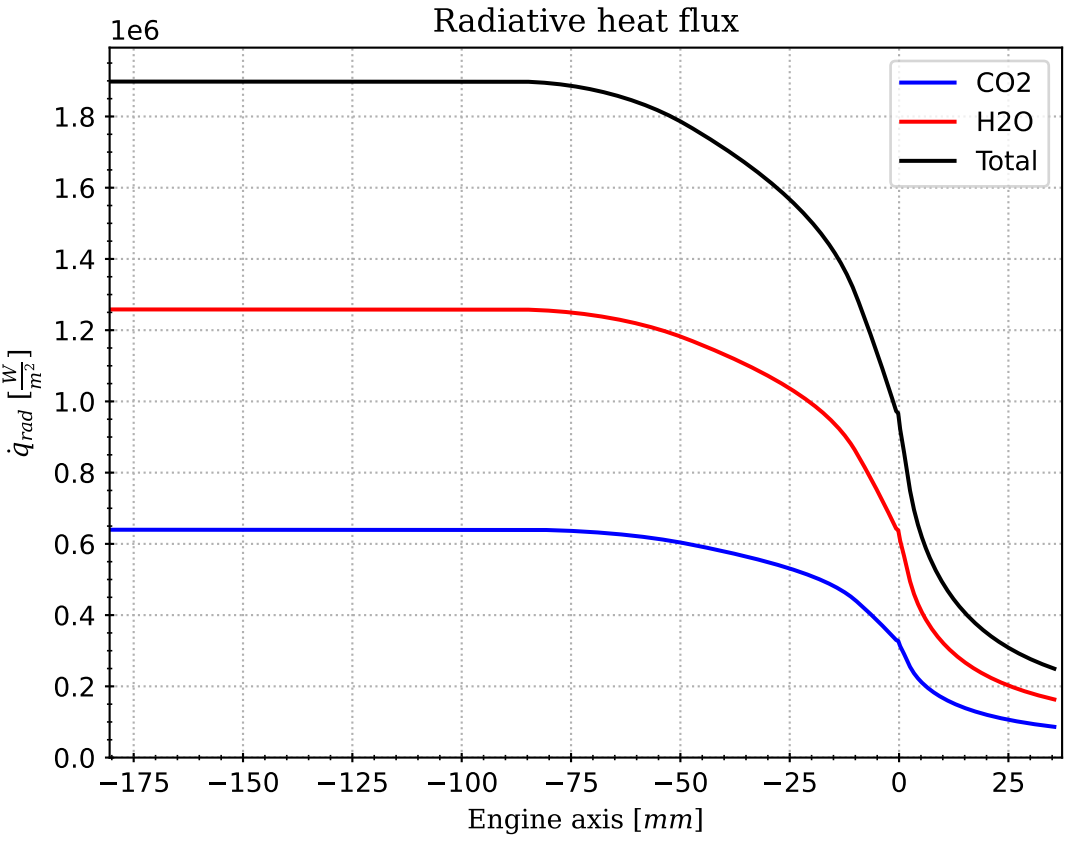

Hot-side convection coefficient h_g
18000
16000
14000
12000
10000
8000
6000
4000
P [Pa]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

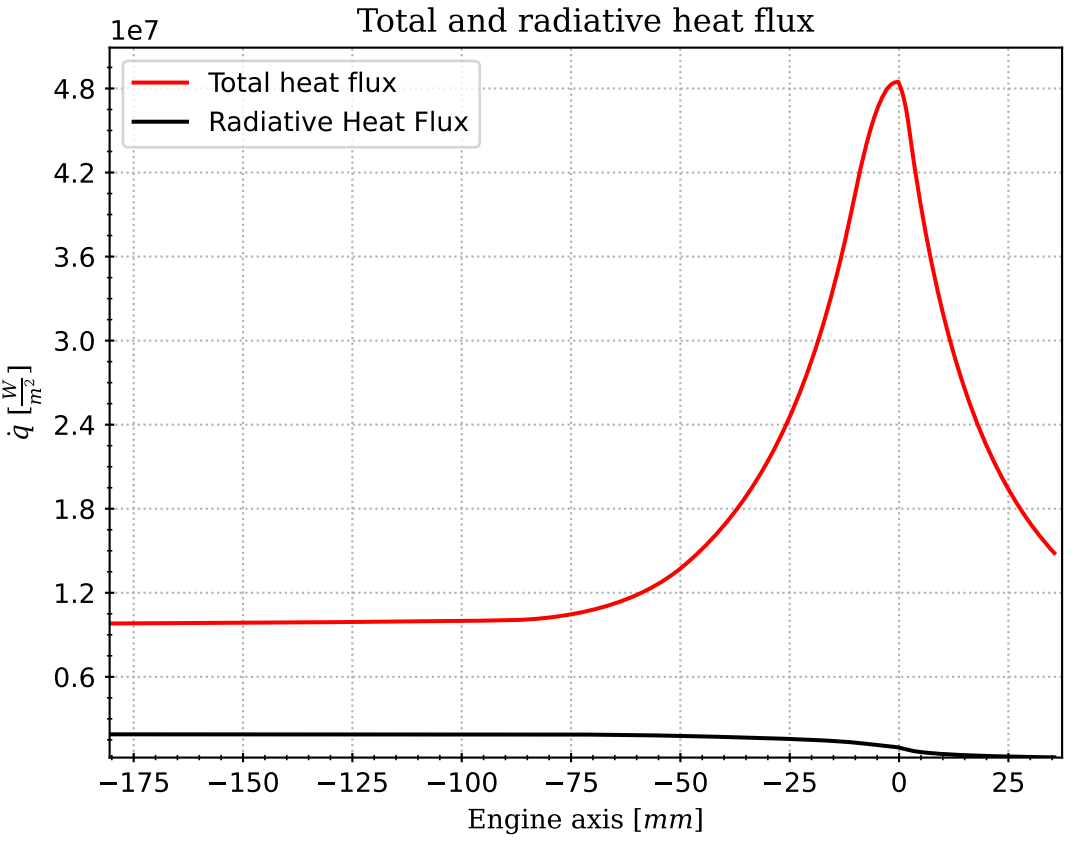

Bartz equation coefficient σ
1.48
1.44
1.40
1.36
1.32
1.28
1.24
σ [-]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

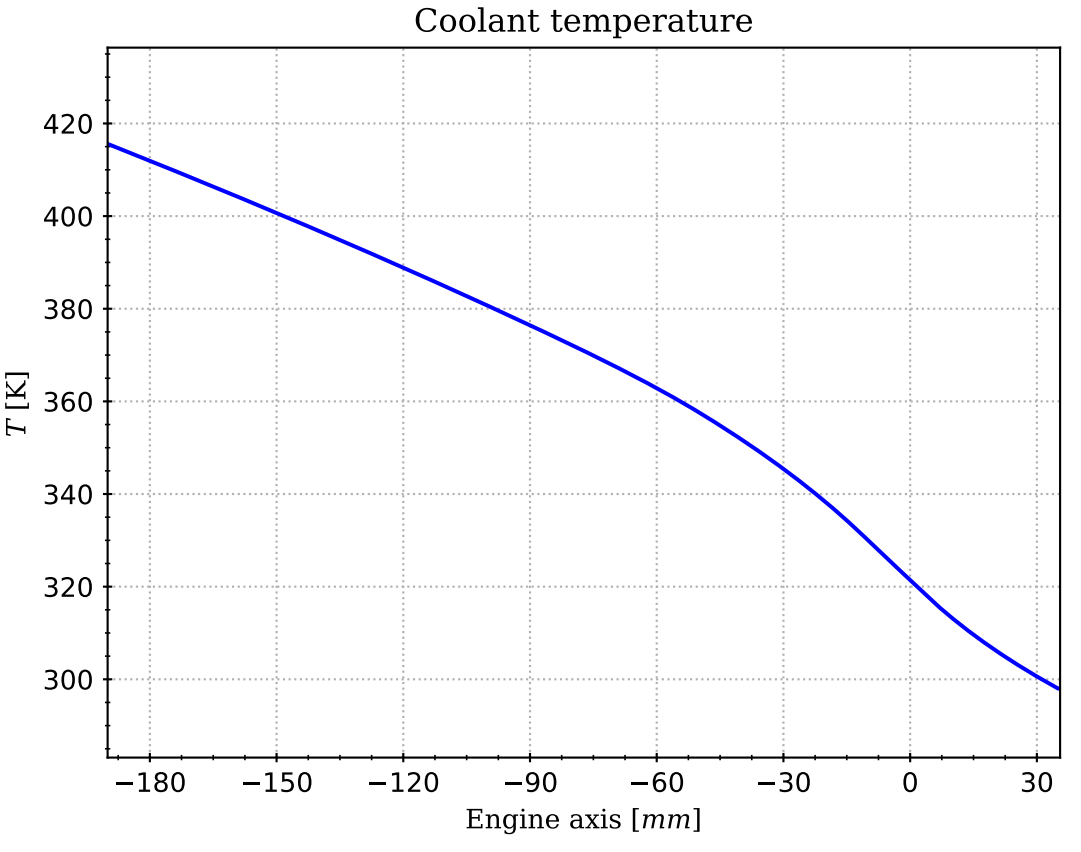

Cold-side convective coefficient h_l
No correction
Correction (Popp & Schmidt)
135000
120000
105000
90000
75000
60000
45000
30000
15000
0
h_l $[\frac{W}{m^2 \cdot K}]$
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis $[mm]$

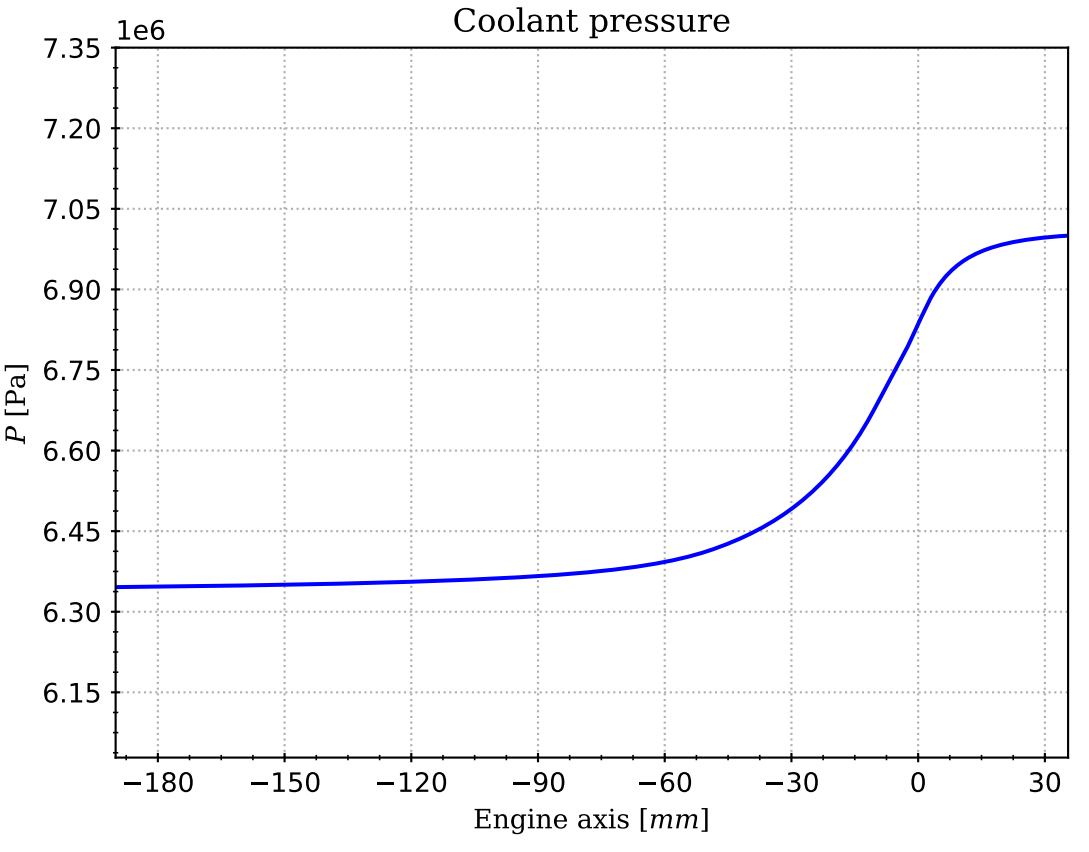

Molar fraction of combustion products
Molar fraction x_i [-]
0.48
0.42
0.36
0.30
0.24
0.18
0.12
0.06
0.00
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]
H_2O
CO_2

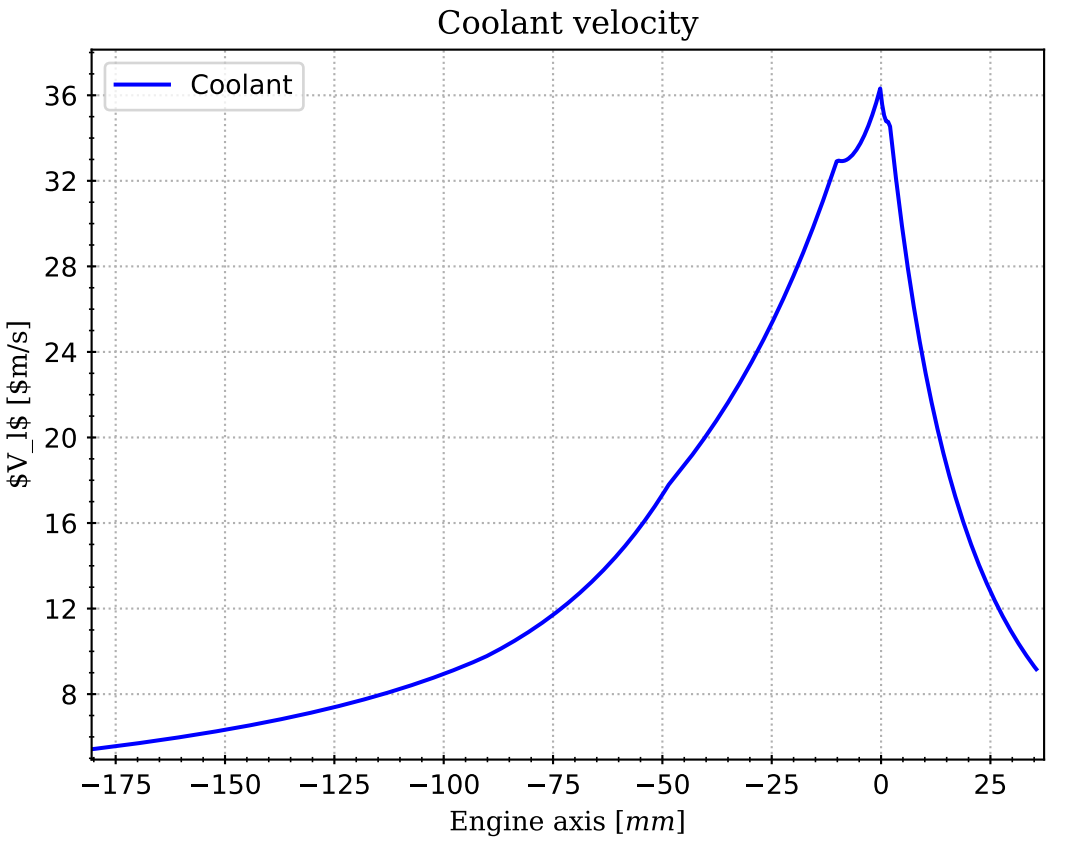

Partial pressure of combustion products
1e6
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Partial pressure p_i [Pa]
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]
H_2O
CO_2

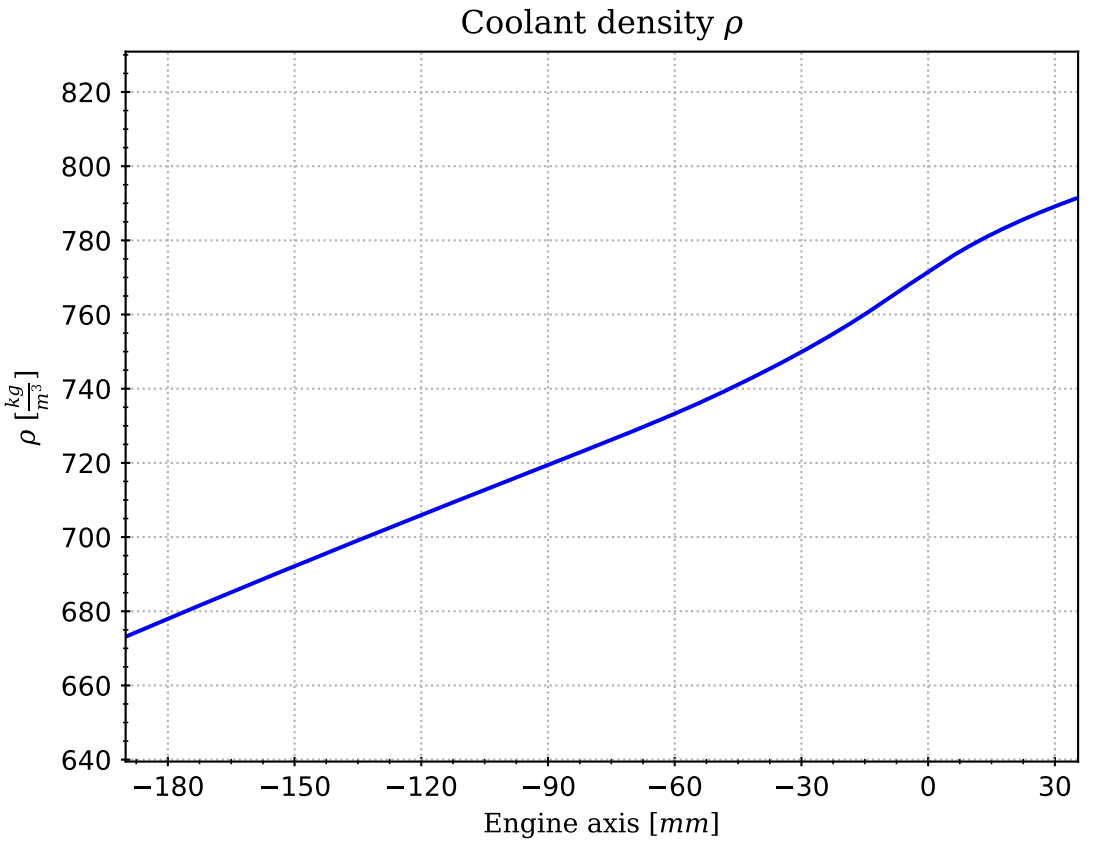

Radiative heat flux
1e6
CO2
H2O
Total
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
$\dot{q}_{rad}$ $[\frac{W}{m^2}]$
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis $[mm]$

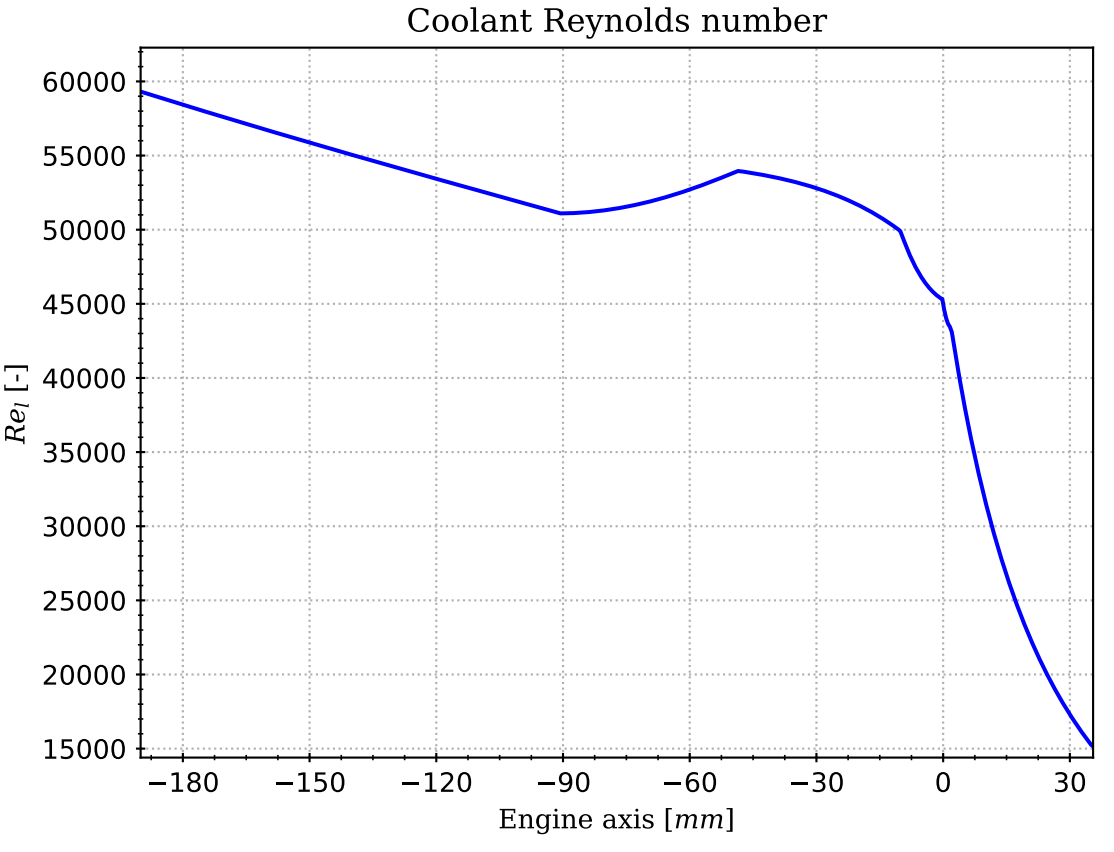

Total and radiative heat flux
1e7
Total heat flux
Radiative Heat Flux
4.8
4.2
3.6
3.0
2.4
1.8
1.2
0.6
$\dot{q}\ [\frac{W}{m^2}]$
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis $[mm]$

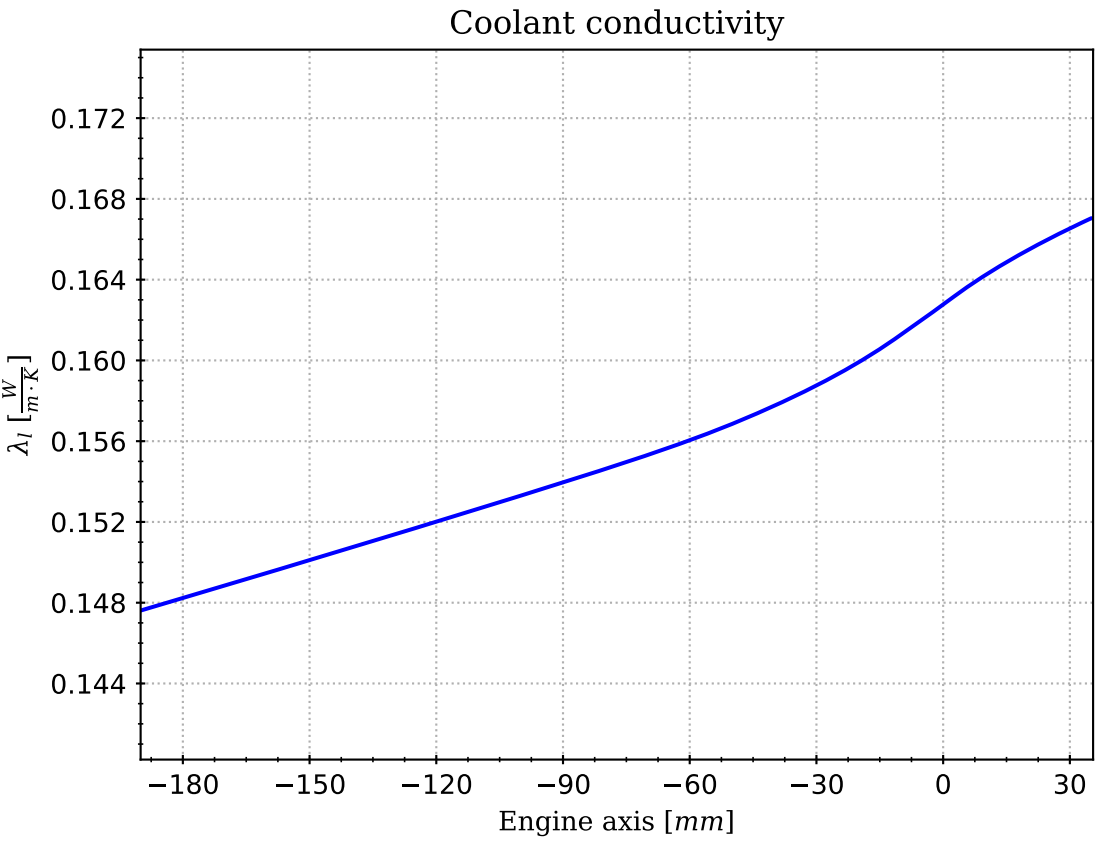

Coolant temperature
420
400
380
360
340
320
300
T [K]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

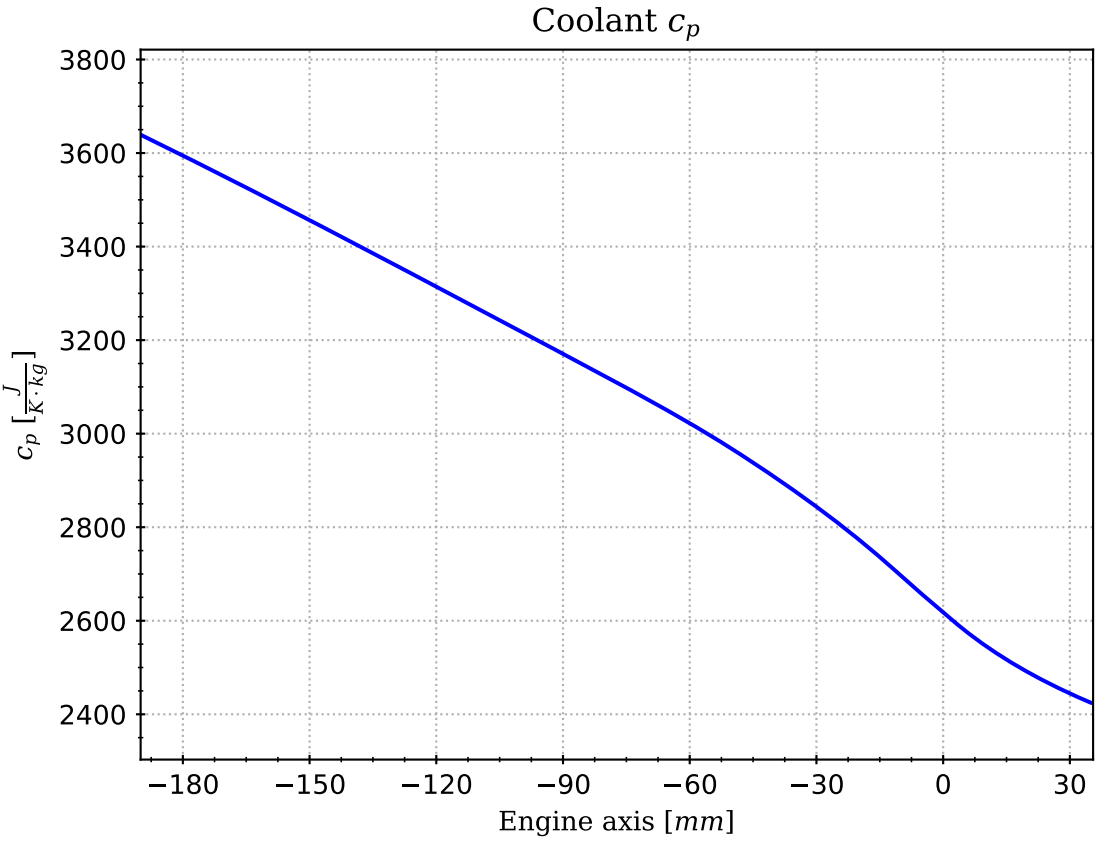

Coolant pressure
1e6
7.35
7.20
7.05
6.90
6.75
6.60
6.45
6.30
6.15
P [Pa]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

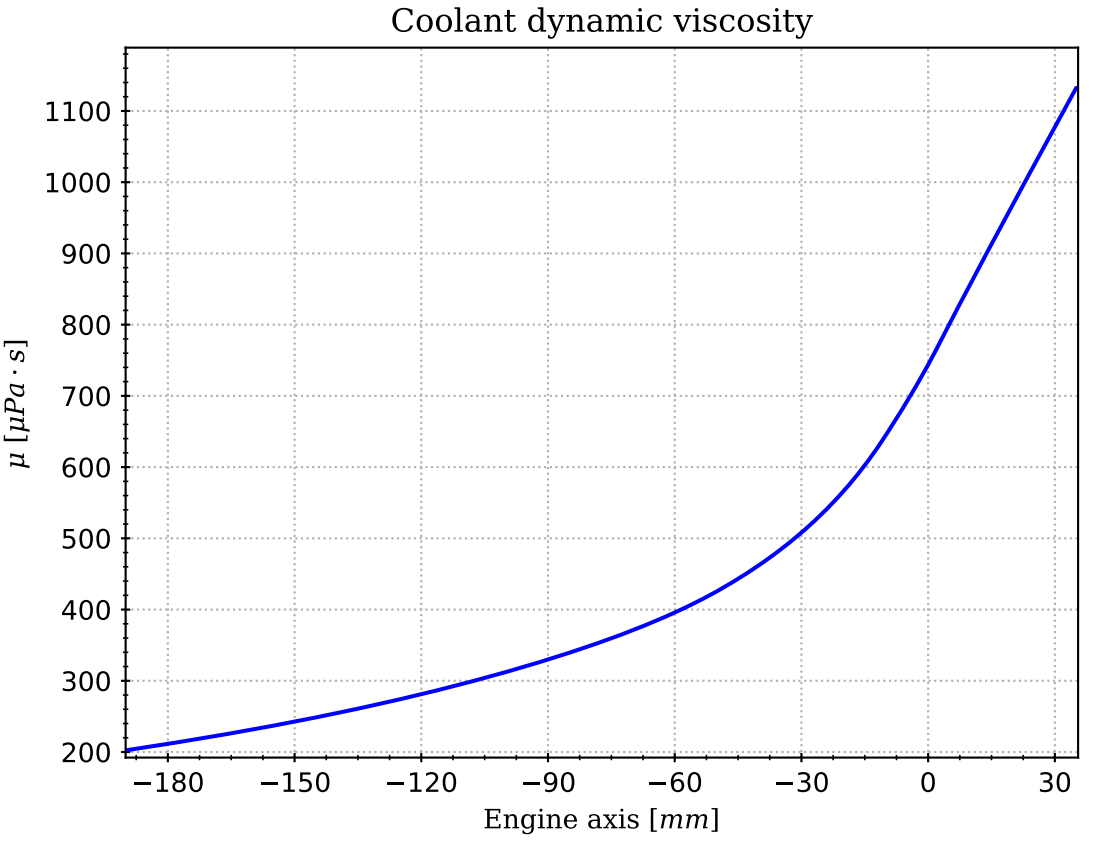

Coolant velocity
Coolant
36
32
28
24
20
16
12
8
V_l [$m/s]
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]

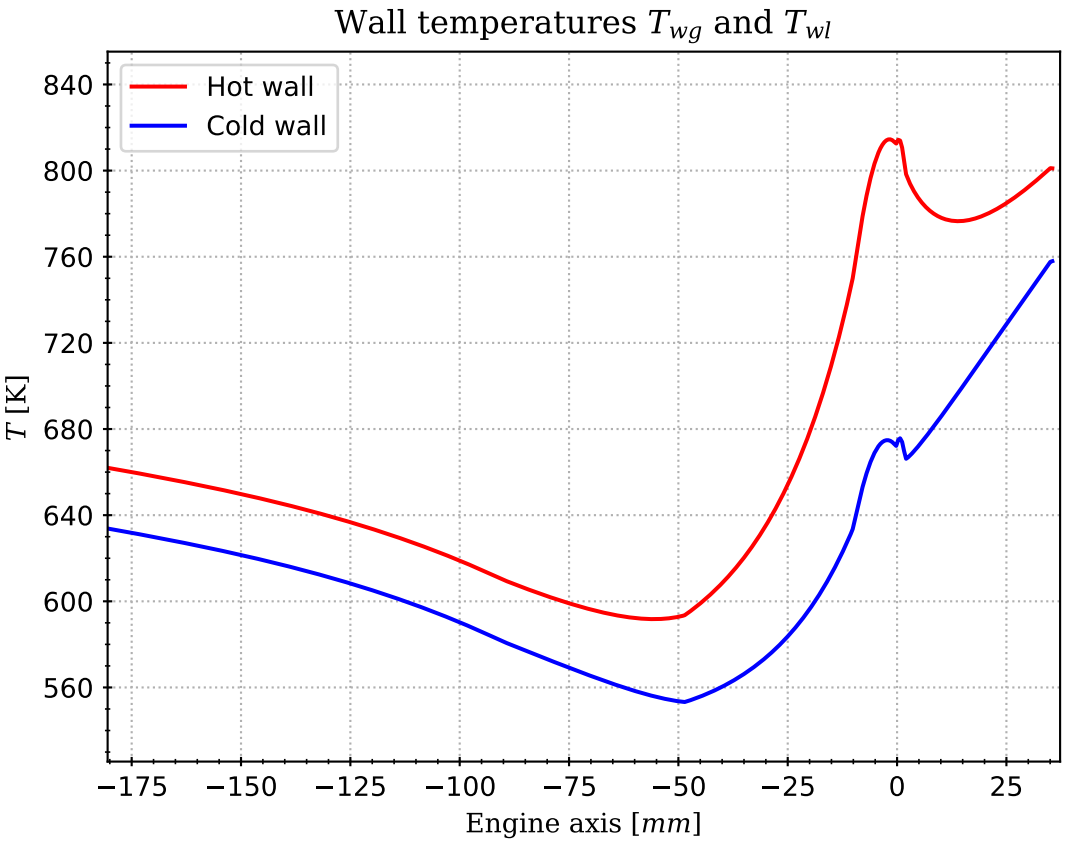

Coolant density ρ
820
800
780
760
740
720
700
680
660
640
$\rho\ [\frac{kg}{m^3}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis $[mm]$

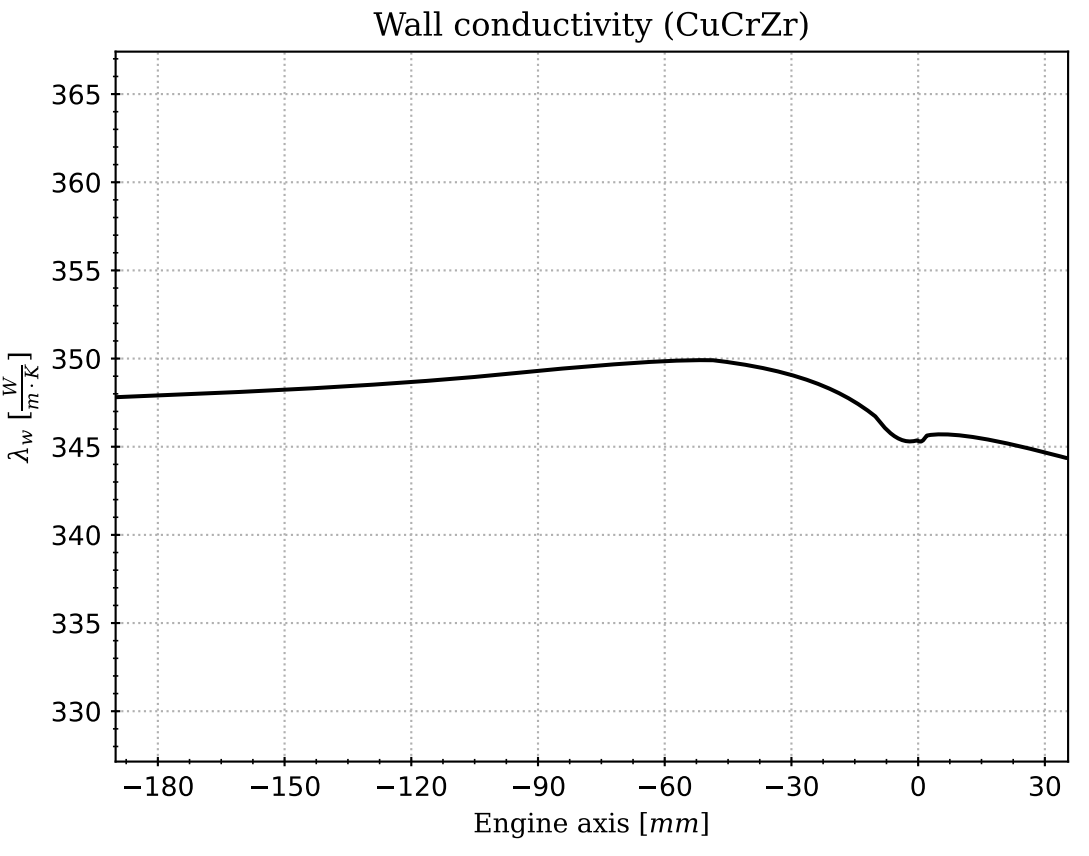

Coolant Reynolds number
60000
55000
50000
45000
40000
35000
30000
25000
20000
15000
Re_l [-]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]


Coolant conductivity
0.172
0.168
0.164
0.160
0.156
0.152
0.148
0.144
λ_l $[\frac{W}{m \cdot K}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis $[mm]$


Coolant c_p
3800
3600
3400
3200
3000
2800
2600
2400
c_p $[\frac{J}{K \cdot kg}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis $[mm]$


Coolant dynamic viscosity
1100
1000
900
800
700
600
500
400
300
200
μ [$\mu Pa \cdot s$]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]


Wall temperatures T_{wg} and T_{wl}
Hot wall
Cold wall
840
800
760
720
680
640
600
560
T [K]
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]


Wall conductivity (CuCrZr)
365
360
355
350
345
340
335
330
λ_w $[\frac{W}{m \cdot K}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis $[mm]$